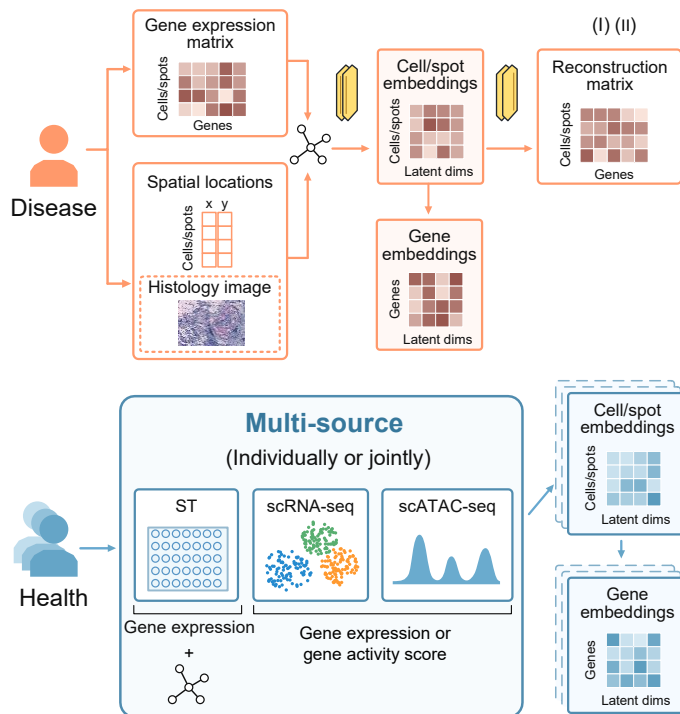
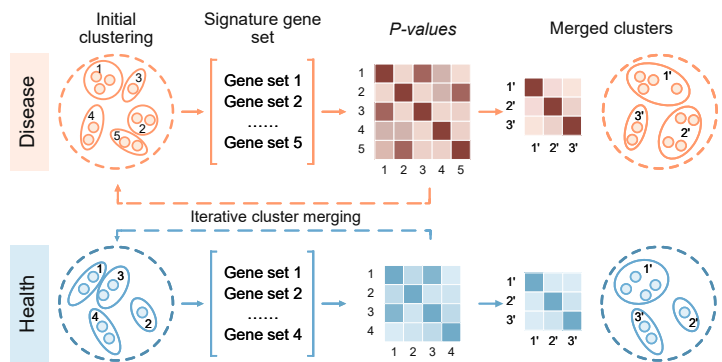


Part 1: Pathological-region detection

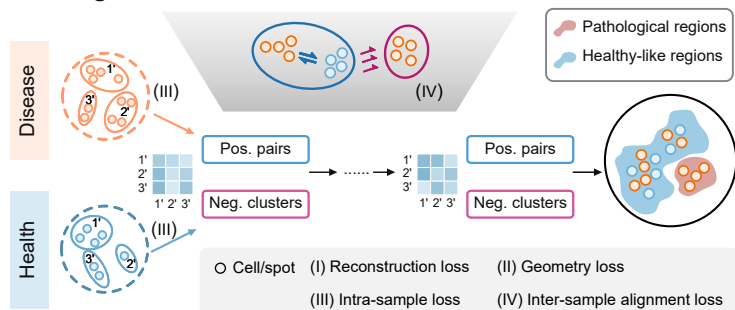
a Co-embedding learning



b Automated clustering



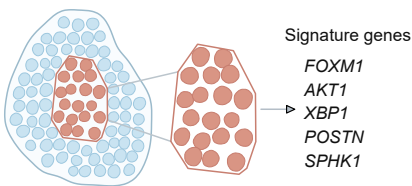
c Integration & Detection



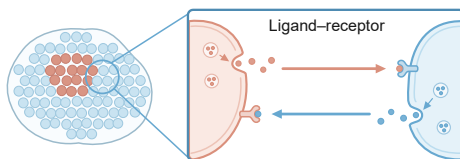
Part 2: Pathological-region characterization

d

Pathological-region signature genes



Pathological-healthy communications



Intra-region heterogeneity

